



ϕ -linear canonical analytic signals



Yan-Na Zhang^{a,b}, Bing-Zhao Li^{a,b,*}

^a School of Mathematics and Statistics, Beijing Institute of Technology, Beijing 100081, China

^b Beijing Key Laboratory on MCAACI, Beijing Institute of Technology, Beijing 100081, China

ARTICLE INFO

Article history:

Received 6 September 2016

Revised 24 August 2017

Accepted 6 September 2017

Available online 8 September 2017

Keywords:

ϕ -linear canonical Hilbert transform

ϕ -linear canonical analytic signal

Linear canonical transform

Single-sideband modulation

ABSTRACT

In this paper, a ϕ -linear canonical Hilbert transform (ϕ -LCHT) is proposed by a linear combination of a given signal and its parameter Hilbert transform (PHT). Herein, a PHT is a generalized HT based on the linear canonical transform (LCT). Some essential properties of a ϕ -LCHT are also derived. According to the derivation of a ϕ -LCHT, four different versions of ϕ -linear canonical analytic signals (ϕ -LCASs) are defined and analyzed in detail. Furthermore, ϕ -LCASs are discussed and compared by some valuable properties such as the LCT spectrum and ϕ -LCHT eigenfunction property. In terms of application, a multi-key single-sideband (SSB) modulation scheme is investigated using one of the ϕ -LCASs proposed in this paper, and its performance with respect to noise and sensitivity to security key perturbations is analyzed.

© 2017 Elsevier B.V. All rights reserved.

1. Introduction

The Hilbert transform (HT) plays an important role in signal processing. It was first introduced into the communication field by Gabor in 1946 [1]. He constructed a complex signal called an analytic signal (AS) by adding a given real signal in quadrature with its HT. One of the most important properties of an AS is that it contains no negative Fourier frequency, which is commonly referred to as the one-sided spectrum. This property is significant in single-sideband (SSB) modulation applications where an AS can reduce the signals transmission bandwidth by half. In addition, an AS is also an eigenfunction of the HT operator. The HT and AS have many interesting and important applications including image edge enhancement [2], instantaneous frequency estimation [3,4], spectral analysis [5], sampling rate conversion [6], and time delay estimation [7,8].

The classical AS is closely related to the Fourier transform (FT) and HT. Since FT analysis is limited for non-stationary signals, more and more novel transformation methods have recently been proposed such as the wavelet transform (WT) [9], fractional Fourier transform (FrFT) [10], and linear canonical transform (LCT) [11,12]. With respect to the aforementioned applications of the AS and HT and the flexibility of these emerging approaches over the FT, a large number of attempts have been made to improve the performances of both the HT and AS by extending them into new transformed domains. In 1998, Zayed combined the HT with the FrFT and proposed the fractional Hilbert transform (FrHT). A generaliza-

tion of the AS was also provided by suppressing the negative frequency of the FrFT signal [13]. However, this extension of the HT and AS lacks the semi-group property of the FrFT. Fu and Li extended the HT into the LCT domain and proposed the parameter Hilbert transform (PHT). In their work, they presented a generalized analytic signal (GAS) by suppressing the negative frequency components in the LCT domain [14]. They also derived a generalized Bedrosian theorem related to the LCT [14,15]. Note however, that the definition of the GAS by Fu et al. has some drawbacks. In particular, for the classical AS, it is sufficient to carry all of the signal information but only with half-bandwidth, mainly due to the fundamental conjugate symmetry property of the FT for a real signal. On the contrary, the LCT for a given real signal lacks this important property. Thus, only by suppressing the negative portion of LCT-transformed signals, the original signal can hardly recovered from the corresponding GAS. To address this drawback, Pei proposed another AS based on the LCT called the canonical analytic signal (CAS) [16]. Moreover, two-dimensional HTs associated with the LCT were proposed by Xu et al. in a different way. They also investigated generalized uncertainty principles related to the HT and LCT [17,18]. Discrete implementations for the HT and its generalized forms are provided in the literature [19–21].

Another way to improve the HT is by adding a rotation parameter in terms of the filter function [22], which is different from transform techniques. It possesses the semi-group property, while fractional analytic signals (FASs) associated with the rotation parameter [23] cannot always contain a one-sided spectrum. On the other hand, the generalized-phase analytic signal (GPAS) was proposed by Venkitaraman et al. based on the generalized-phase Hilbert transform (GPHT) [24]. The phase of the GPHT in

* Corresponding author.

E-mail address: li_bingzhao@bit.edu.cn (B.-Z. Li).

the Fourier domain is not constant, but varies over the Fourier frequency. Both the HT and its extensions are valuable components in AS (or generalized AS) construction and play crucial roles in many practical applications such as SSB modulation and image enhancement. In consideration of the drawbacks of the existing methods, it is necessary to develop a better definition. The main goal of this paper is to propose a new definition of the HT, combining the rotation parameter with the LCT; specifically, it is a rotation-weighted original signal and its generalized HT based on the LCT called as ϕ -linear canonical Hilbert transform (ϕ -LCHT). It can involve most extensions of the HT as its special cases and satisfy specific properties. Some important properties of the ϕ -LCHT are discussed. Based on the rotation parameter and LCT, four different extensions of the AS are proposed and analyzed, which we refer to as ϕ -linear canonical analytic signals (ϕ -LCASs). In addition, one of ϕ -LCASs is successfully applied to a SSB modulation application and the performance of experimental results is discussed in detail.

The remainder of this paper is organized as follows. Section 2 provides some basic knowledge of related work on the HT and AS. In Section 3, the definition and properties of the ϕ -LCHT are proposed. In addition, derivations of the ϕ -LCHT are analyzed from two different views as well. Four ϕ -LCASs are presented in Section 4. The differences between these new analytic signals are compared in terms of the LCT spectrum characteristic and eigenfunction property of the ϕ -LCHT. In order to better understand ϕ -LCASs, Section 5 summarizes their merits and conducts further investigations using the Wigner–Ville distribution (WVD) in the time-frequency plane. Finally, applications and conclusions are stated in Sections 6 and 7, respectively.

2. Preliminaries

2.1. Hilbert transform and analytic signal

The HT of a real-valued signal $s(t)$ is defined by [1]

$$\tilde{s}(t) = \mathcal{H}[s](t) = p.v. \frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{s(\tau)}{t - \tau} d\tau = (s * h)(t), \quad (1)$$

where $\mathcal{H}$ is the HT operator, p.v. denotes the Cauchy principal value integral, $h(t) = 1/\pi t$, and $*$ is the convolution operator. The corresponding AS of $s(t)$ is

$$z(t) = s(t) + j\tilde{s}(t) = s(t) + j\mathcal{H}[s](t). \quad (2)$$

In the Fourier domain, the AS is represented as

$$Z(w) = (1 + \text{sgn}(w))S(w) = 2U(w)S(w), \quad (3)$$

where w is the Fourier frequency, $\text{sgn}(\cdot)$ denotes the signum signal, $U(\cdot)$ is the unit-step signal, and $Z(w)$, $S(w)$ represents the FT of $z(t)$, $s(t)$, respectively. Eq. (3) shows that the AS has only a one-sided spectrum in the Fourier domain, which plays an important role in wide practical applications such as SSB modulation and filter design.

2.2. Fractional Hilbert transform and analytic signals

Because the HT is orthogonal to the original signal, it lacks some effectiveness in practical applications. For example, it cannot be used to selectively enhance edges during image processing [2]. Thus, it has been improved by researchers by adding a rotation parameter ϕ . The fractional Hilbert transform (FHT) is defined as a linear combination of $s(t)$ and $\tilde{s}(t)$ [23]:

$$\tilde{s}_\phi(t) = \mathcal{H}_\phi[s](t) = \cos \phi s(t) + \sin \phi \tilde{s}(t), \quad (4)$$

where $\mathcal{H}_\phi$ denotes the FHT operator. Clearly, when $\phi = \pi/2$, $\tilde{s}_\phi(t)$ yields $\tilde{s}(t)$.

According to parameter ϕ , three different fractional analytic signals (FASs) were proposed by Cusmariu [23]:

$$\begin{aligned} z1_\phi(t) &= \mathcal{H}_\phi[z](t) \\ z2_\phi(t) &= s(t) + j\mathcal{H}_\phi[s](t) \\ z3_\phi(t) &= \cos \phi s(t) + j \sin \phi \mathcal{H}[s](t). \end{aligned} \quad (5)$$

Some properties of the aforementioned FASs in Fourier domain are discussed in detail such as instantaneous frequency, instantaneous amplitude, and instantaneous phase [23].

2.3. Parameter Hilbert transform and generalized analytic signal

Another method based on a transform technique to improve the effectiveness of the HT and AS is introduced in this subsection. Different from the FHT and FASs added to a free rotation parameter ϕ , the parameter Hilbert transform (PHT) and generalized analytic signal (GAS) were proposed by Fu and Li [14], which are closely related to a generalized Fourier transform, i.e., the LCT (see Section 2.4).

As the generalizations of HT and AS associated with the LCT, PHT and GAS are defined by [14]

$$\tilde{s}_A(t) = \mathcal{H}_A[s](t) = \frac{e^{-j\frac{a}{2b}t^2}}{\pi} \int_{-\infty}^{+\infty} \frac{s(\tau)e^{j\frac{a}{2b}\tau^2}}{t - \tau} d\tau \quad (6)$$

and

$$z_A(t) = s(t) + j\tilde{s}_A(t) = s(t) + j\mathcal{H}_A[s](t), \quad (7)$$

where $\mathcal{H}_A$ is the PHT operator. Eqs. (6) and (7) reduce to the classical form, HT and AS, respectively, under specific parameters since the LCT equals the FT with $A = (0, 1; -1, 0)$. Similarly, the PHT and GAS reduces to the FrHT and FrAS proposed by Zayed, respectively, with $A = (\cos \alpha, \sin \alpha; -\sin \alpha, \cos \alpha)$ [13].

The LCT of the PHT and GAS when $b > 0$ is

$$\mathcal{L}_{\tilde{s}_A}^A(u) = -j\text{sgn}(u)S^A(u) \quad (8)$$

and

$$\mathcal{L}_{z_A}^A(u) = (1 + \text{sgn}(u))S^A(u) = 2U(u)S^A(u), \quad (9)$$

respectively. From Eq. (9), the GAS is obtained by suppressing the negative frequency content of the LCT; that is, it contains a one-sided spectrum in the LCT domain.

2.4. The linear canonical transform

The LCT of a signal $s(t)$ with parameter matrix A is defined by [11,12]

$$S^A(u) = \mathcal{L}_s^A(u) = \begin{cases} \int_{-\infty}^{+\infty} s(t)K_A(u, t)dt, & b \neq 0 \\ \sqrt{d}e^{j(cd/2)u^2}S(du), & b = 0 \end{cases}, \quad (10)$$

where $K_A(u, t)$ denotes the LCT kernel and is given by

$$K_A(u, t) = \frac{1}{\sqrt{j2\pi b}} e^{j\left(\frac{du^2}{2b} - \frac{ut}{b} + \frac{at^2}{2b}\right)} \quad (11)$$

with parameter matrix $A = (a, b; c, d)$, where a, b, c , and d are real numbers satisfying $ad - bc = 1$, i.e., $\det(A) = 1$.

The inverse linear canonical transform (ILCT) is another LCT whose parameter matrix equals the inverse matrix of the LCT with A . It can be rewritten as [11,12]

$$s(t) = \mathcal{L}_{S^A}^{A^{-1}}(t) = \begin{cases} \int_{-\infty}^{+\infty} S^A(u)K_{A^{-1}}(t, u)du, & b \neq 0 \\ \sqrt{a}e^{j(-ca/2)t^2}S^A(at), & b = 0 \end{cases}, \quad (12)$$

where $A^{-1} = (d, -b; -c, a)$ is the inverse matrix of A .

The LCT is particularly powerful in various applications such as solving differential equations, optical systems, quantum mechanics and image processing [25,26]. Many well-known signal processing operations, involving the FT, FrFT, Fresnel transform, Lorentz transform, and scaling operation are special cases of the LCT [27].

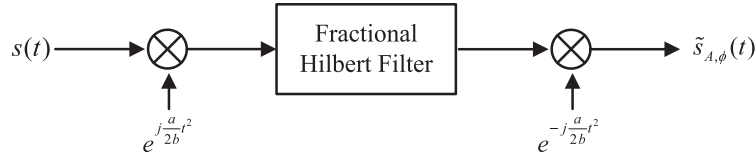


Fig. 1. Construction of the ϕ -linear canonical Hilbert transform (ϕ -LCHT).

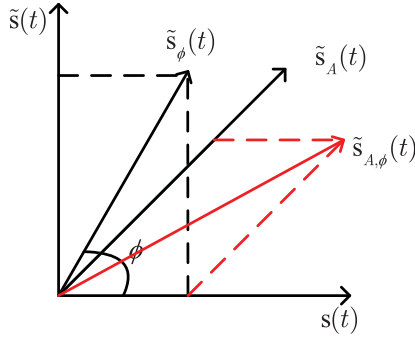


Fig. 2. Geometric interpretation of the ϕ -LCHT.

3. ϕ -linear canonical Hilbert transform

To further improve the performance of the HT, extensions of the HT based on a rotation parameter and the LCT are proposed in this section; specifically, we propose the ϕ -LCHT. The ϕ -LCHT is defined by combining the rotation parameter and LCT with the classical HT. Because of the above parameters, the ϕ -LCHT has an advantage in that it contains many more degrees of freedom than that of the FHT (only with the rotation parameter) and PHT (only related to LCT parameters).

3.1. Definition of the ϕ -linear canonical Hilbert transform

Definition 1. The ϕ -LCHT of a real-valued signal $s(t)$ is defined by

$$\tilde{s}_{A,\phi}(t) = \mathcal{H}_{A,\phi}[s](t) = \cos \phi s(t) + \sin \phi \tilde{s}_A(t), \quad (13)$$

where $A = (a, b; c, d)$ is an LCT parameter matrix satisfying $ad - bc = 1$, $b \neq 0$. ϕ denotes the rotation parameter, $-\pi/2 \leq \phi \leq \pi/2$, $\mathcal{H}_{A,\phi}$ is a ϕ -LCHT operator, and $\tilde{s}_A(t)$ stands for the PHT of $s(t)$.

From Definition 1, the ϕ -LCHT can be rewritten in convolution form as

$$\tilde{s}_{A,\phi}(t) = e^{-j\frac{a}{2b}t^2} \left[\left(s(t) e^{j\frac{a}{2b}t^2} \right) * h_\phi(t) \right], \quad (14)$$

where $h_\phi(t) = \cos \phi \delta(t) + \sin \phi \frac{1}{\pi t}$. Equivalently, computing the ϕ -LCHT of a signal $s(t)$ is equivalent to multiplying it by a chirp $e^{j\frac{a}{2b}t^2}$, then passing the product through a fractional Hilbert filter, and finally multiplying the output by another chirp $e^{-j\frac{a}{2b}t^2}$. This process is shown in Fig. 1. In addition, the former HT, FHT, and PHT are all special cases of the ϕ -LCHT when the following parameters are specified:

- $\phi = 0$, $\mathcal{H}_{A,\phi}[s](t) = s(t)$. The ϕ -LCHT reduces to identity $\mathcal{I}$;
- $\phi = \pi/2$, $\mathcal{H}_{A,\phi}[s](t) = \mathcal{H}_A[s](t)$. The ϕ -LCHT coincides with the PHT (includes the HT and FrHT as special cases);
- $A = (0, 1; -1, 0)$, $\mathcal{H}_{A,\phi}[s](t) = \mathcal{H}_\phi[s](t)$. The ϕ -LCHT coincides with the FHT.

3.2. Properties of the ϕ -linear canonical Hilbert transform

To better understand and use this new definition, we consider the following properties of ϕ -LCHTs.

Property 1. Linearity: For any constants k_1 and k_2 and signals $f(t)$ and $g(t)$, $\mathcal{H}_{A,\phi}[k_1 f + k_2 g](t) = k_1 \mathcal{H}_{A,\phi}[f](t) + k_2 \mathcal{H}_{A,\phi}[g](t)$.

Property 2. Semi-group: For any rotation parameters ϕ_1 and ϕ_2 , it is true that $\mathcal{H}_{A,\phi_2} \mathcal{H}_{A,\phi_1}[s](t) = \mathcal{H}_{A,\phi_2+\phi_1}[s](t)$.

Proof. By utilizing the LCT of the PHT in Eq. (8), the LCT of $\mathcal{H}_A[\tilde{s}_A](t)$ is

$$\mathcal{L}_{\mathcal{H}_A[\tilde{s}_A]}(u) = -j \operatorname{sgn}(u) \mathcal{L}_{\tilde{s}_A}^A(u) = -S^A(u). \quad (15)$$

The ILCT is taken to both sides of the above equation, which yields

$$\mathcal{H}_A^2[s](t) = -s(t). \quad (16)$$

Based on Eq. (16),

$$\begin{aligned} \mathcal{H}_{A,\phi_2} \mathcal{H}_{A,\phi_1}[s](t) &= \cos \phi_2 \mathcal{H}_{A,\phi_1}[s](t) + \sin \phi_2 \mathcal{H}_A \mathcal{H}_{A,\phi_1}[s](t) \\ &= \cos \phi_2 \cos \phi_1 s(t) + \cos \phi_2 \sin \phi_1 \mathcal{H}_A[s](t) \\ &\quad + \sin \phi_2 \cos \phi_1 \mathcal{H}_A[s](t) + \sin \phi_2 \sin \phi_1 \mathcal{H}_A^2[s](t) \\ &= \cos \phi_2 \cos \phi_1 s(t) + \cos \phi_2 \sin \phi_1 \mathcal{H}_A[s](t) \\ &\quad + \sin \phi_2 \cos \phi_1 \mathcal{H}_A[s](t) - \sin \phi_2 \sin \phi_1 s(t) \\ &= \cos(\phi_1 + \phi_2) s(t) + \sin(\phi_1 + \phi_2) \mathcal{H}_A[s](t) \\ &= \mathcal{H}_{A,\phi_2+\phi_1}[s](t). \end{aligned} \quad (17)$$

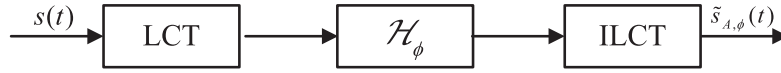
The proof is completed. $\square$

Property 3. Scale-change: If m and n are any constants, then $\mathcal{H}_{A,\phi}[f(mx+n)](t) = \cos \phi f(mt+n) + \sin \phi e^{j\frac{a_1}{b_1}n(mt+n)} \mathcal{H}_{A_1}[f(x)e^{-j\frac{a_1}{b_1}nx}](mt+n)$, where $\frac{a_1}{b_1} = \frac{1}{m^2} \frac{a}{b}$. Specifically, when $n = 0$, it yields $\mathcal{H}_{A,\phi}[f(mx)](t) = \mathcal{H}_{A_1,\phi}[f](mt)$.

Proof. Let $g(t) = f(mt+n)$. The scale-change property of the PHT is given first. It is

$$\begin{aligned} \mathcal{H}_A[g](t) &= \frac{e^{-j\frac{a}{2b}t^2}}{\pi} \int_{-\infty}^{+\infty} \frac{g(\tau) e^{j\frac{a}{2b}\tau^2}}{t-\tau} d\tau \\ &\stackrel{\text{Let } \lambda=mx+n}{=} \frac{e^{-j\frac{a}{2b}t^2}}{\pi} \int_{-\infty}^{+\infty} \frac{f(\lambda) e^{j\frac{a(\lambda-n)^2}{2bm^2}}}{t-\frac{\lambda-n}{m}} \frac{1}{m} d\lambda \\ &= \frac{e^{-j\frac{a}{2b}t^2}}{\pi} \cdot e^{j\frac{an^2}{2bm^2}} \int_{-\infty}^{+\infty} \frac{f(\lambda) e^{-j\frac{a\lambda^2}{2bm^2}}}{mt+n-\lambda} \cdot e^{j\frac{a\lambda^2}{2bm^2}} d\lambda \\ &= e^{-j\frac{a(mt+n)^2}{2bm^2}} \cdot e^{j\frac{an(mt+n)}{bm^2}} \int_{-\infty}^{+\infty} \frac{f(\lambda) e^{-j\frac{a\lambda^2}{2bm^2}}}{\pi(mt+n-\lambda)} e^{j\frac{a\lambda^2}{2bm^2}} d\lambda \\ &= e^{-j\frac{a_1}{2b_1}(mt+n)^2} \cdot e^{j\frac{a_1 n(mt+n)}{b_1}} \int_{-\infty}^{+\infty} \frac{f(\lambda) e^{-j\frac{a_1 \lambda^2}{2b_1}}}{\pi(mt+n-\lambda)} e^{j\frac{a_1 \lambda^2}{2b_1}} d\lambda \\ &= e^{j\frac{a_1}{b_1}n(mt+n)} \mathcal{H}_{A_1} \left[f(x) e^{-j\frac{a_1}{b_1}nx} \right] (mt+n), \end{aligned} \quad (18)$$

where $\frac{a_1}{b_1} = \frac{a}{bm^2}$. When $n = 0$, it is easy to verify that $\mathcal{H}_A[f(mx)](t) = \mathcal{H}_{A_1}[f](mt)$. Based on the scale-change property of the PHT and the linearity of the ϕ -LCHT, the associated scale-change property of the ϕ -LCHT is obtained. $\square$

Fig. 3. Filter interpretation of the ϕ -LCHT.

Property 4. Inversion property: $\mathcal{H}_{A,\alpha}^{-1}[s](t) = \mathcal{H}_{A,-\alpha}[s](t)$.

Proof. Using **Property 2**, let $\phi_2 = -\phi_1$. Then

$$\mathcal{H}_{A,-\phi_1}\mathcal{H}_{A,\phi_1}[s](t) = \mathcal{H}_{A,0}[s](t) = s(t). \quad (19)$$

□

3.3. Interpretation of the ϕ -linear canonical Hilbert transform

In this subsection, two ways of understanding the ϕ -LCHT are provided. One is given by a geometric interpretation treating signals as vectors, and the other is from a filter interpretation.

3.3.1. Geometric interpretation of the ϕ -linear canonical Hilbert transform

The geometric interpretation of the ϕ -LCHT is provided by treating signals as vectors. Since the HT is orthogonal to the original signal, $\tilde{s}(t)$ and $s(t)$ are bases in the rectangular coordinate system. The FHT is a rotation combination of $\tilde{s}(t)$ and $s(t)$. The PHT is not always orthogonal to the original signal owing to the existence of extra parameters, i.e., parameter matrix A associated with the LCT. According to the definition of the ϕ -LCHT, which is a rotation combination of $\tilde{s}_A(t)$ and $s(t)$, the ϕ -LCHT is constructed in an affine coordinate system (treating $\tilde{s}_A(t)$ and $s(t)$ as a basis). It is a generalization of the HT, FHT, and PHT. These relationships are shown in Fig. 2.

3.3.2. Filter interpretation of the ϕ -linear canonical Hilbert transform

Next, the ϕ -LCHT is explained from an LCT filter. In this view, the ϕ -LCHT is equivalent to the flow chart in Fig. 3. Specifically, the ϕ -LCHT of a given signal $s(t)$ is derived from the following process: first, obtain the LCT of an input signal $s(t)$ with parameter matrix A , and then transmit it through an FHT filter with rotation parameter ϕ in the LCT domain; finally, the signal is transformed with the inverse matrix of A to obtain the output signal in the time domain, and the output signal is exactly the ϕ -LCHT of $s(t)$.

That is to say,

$$\begin{aligned} \mathcal{H}_{A,\phi}[s](t) &= \mathcal{L}^{A^{-1}} \mathcal{H}_{\phi} \mathcal{L}_s^A(t) \\ &= \int_{-\infty}^{+\infty} (\cos \phi - j \sin \phi \operatorname{sgn}(u)) S^A(u) K_{A^{-1}}(u, t) du \\ &= \int_{-\infty}^{+\infty} (\cos \phi - j \sin \phi \operatorname{sgn}(u)) \\ &\quad \left[\int_{-\infty}^{+\infty} s(\tau) K_A(u, \tau) d\tau \right] K_{A^{-1}}(u, t) du \\ &= \frac{e^{-j\frac{at^2}{2b}}}{2\pi b} \int_{-\infty}^{+\infty} s(\tau) e^{j\frac{a\tau^2}{2b}} d\tau \\ &\quad \int_{-\infty}^{+\infty} (\cos \phi - j \sin \phi \operatorname{sgn}(u)) e^{-j\frac{u(\tau-t)}{b}} du. \end{aligned} \quad (20)$$

According to

$$\begin{aligned} &\frac{1}{2\pi} \int_{-\infty}^{+\infty} (\cos \phi - j \sin \phi \operatorname{sgn}(u)) e^{-jut} dt \\ &= \cos \phi \delta(t) + \sin \phi \frac{1}{\pi t}, \end{aligned} \quad (21)$$

we derive

$$\begin{aligned} \mathcal{H}_{A,\phi}[s](t) &= \frac{e^{-j\frac{at^2}{2b}}}{b} \int_{-\infty}^{+\infty} s(\tau) e^{j\frac{a\tau^2}{2b}} \\ &\quad \times \left(\cos \phi \delta\left(\frac{\tau-t}{b}\right) + \sin \phi \frac{b}{\pi(\tau-t)} \right) d\tau. \end{aligned} \quad (22)$$

Let

$$\mathcal{H}_{A,\phi}[s](t) = J_1(t) + J_2(t), \quad (23)$$

where

$$\begin{aligned} J_1(t) &= \frac{1}{b} e^{-j\frac{at^2}{2b}} \int_{-\infty}^{+\infty} s(\tau) e^{j\frac{a\tau^2}{2b}} \cos \phi \delta\left(\frac{\tau-t}{b}\right) d\tau \\ &= e^{-j\frac{at^2}{2b}} s(t) e^{j\frac{a\tau^2}{2b}} \cos \phi \\ &= \cos \phi s(t) \end{aligned} \quad (24)$$

and

$$\begin{aligned} J_2(t) &= \frac{1}{b} e^{-j\frac{at^2}{2b}} \int_{-\infty}^{+\infty} s(\tau) e^{j\frac{a\tau^2}{2b}} \sin \phi \frac{b}{\pi(\tau-t)} d\tau \\ &= \sin \phi e^{-j\frac{at^2}{2b}} \int_{-\infty}^{+\infty} \frac{s(\tau) e^{j\frac{a\tau^2}{2b}}}{\pi(\tau-t)} d\tau \\ &= \sin \phi \mathcal{H}_A[s](t). \end{aligned} \quad (25)$$

Here, $A^{-1} = (d, -b; -c, a)$ is the inverse matrix of A .

4. ϕ -linear canonical analytic signals

With the rapid development of the HT in various scientific fields such as optics, image processing, and communications, it is significant to further research related generalized analytic signals. Based on the derivation of the ϕ -LCHT, we proposed some new generalized linear canonical analytic signals, called ϕ -LCASs. Specifically, they are

$$\begin{aligned} z1_{A,\phi}(t) &= e^{-j\frac{at^2}{2b}} \mathcal{H}_{\phi}[z](t) \\ z2_{A,\phi}(t) &= s(t) + e^{j(\pi-\phi)} \mathcal{H}_{A,\phi}[s](t) \\ z3_{A,\phi}(t) &= e^{-j\frac{at^2}{2b}} [s(t) + e^{j(\pi-\phi)} \mathcal{H}_{\phi}[s](t)] \\ z4_{A,\phi}(t) &= \cos \phi s(t) + j \sin \phi \mathcal{H}_A[s](t). \end{aligned} \quad (26)$$

The first ϕ -LCAS is given by a chirp signal multiplied by $z1_{\phi}(t) = \mathcal{H}_{\phi}[z](t)$ in Eq. (5). The second option replaces the HT by the ϕ -LCHT, instead of j of $e^{j(\pi-\phi)}$ in terms of the AS. The third option is a chirp signal multiplied by $z2_{\phi}(t)$; the difference is j substituted by $e^{j(\pi-\phi)}$ in $z2_{\phi}(t)$ as well. The fourth ϕ -LCAS is a geometric meaning by a rotation of ϕ , which is a linear combination of the original signal $s(t)$ and its PHT.

For the AS, it is well-known that the one-sided spectrum property is vital in SSB modulation applications. Besides, the AS is an eigenfunction of the HT operator, that is, $\mathcal{H}[z](t) = -jz(t)$, with eigenvalue $-j$ [23]. Next, the LCT spectrum of ϕ -LCASs and the eigenfunction property of the ϕ -LCHT are discussed to verify whether the one-sided spectrum property and eigenfunction property are preserved. Here, the eigenfunction property of the ϕ -LCHT operator is $\mathcal{H}_{A,\phi}[f](t) = \lambda f(t)$, where $f(t)$ is the eigenfunction of the ϕ -LCHT, and λ is the corresponding eigenvalue.

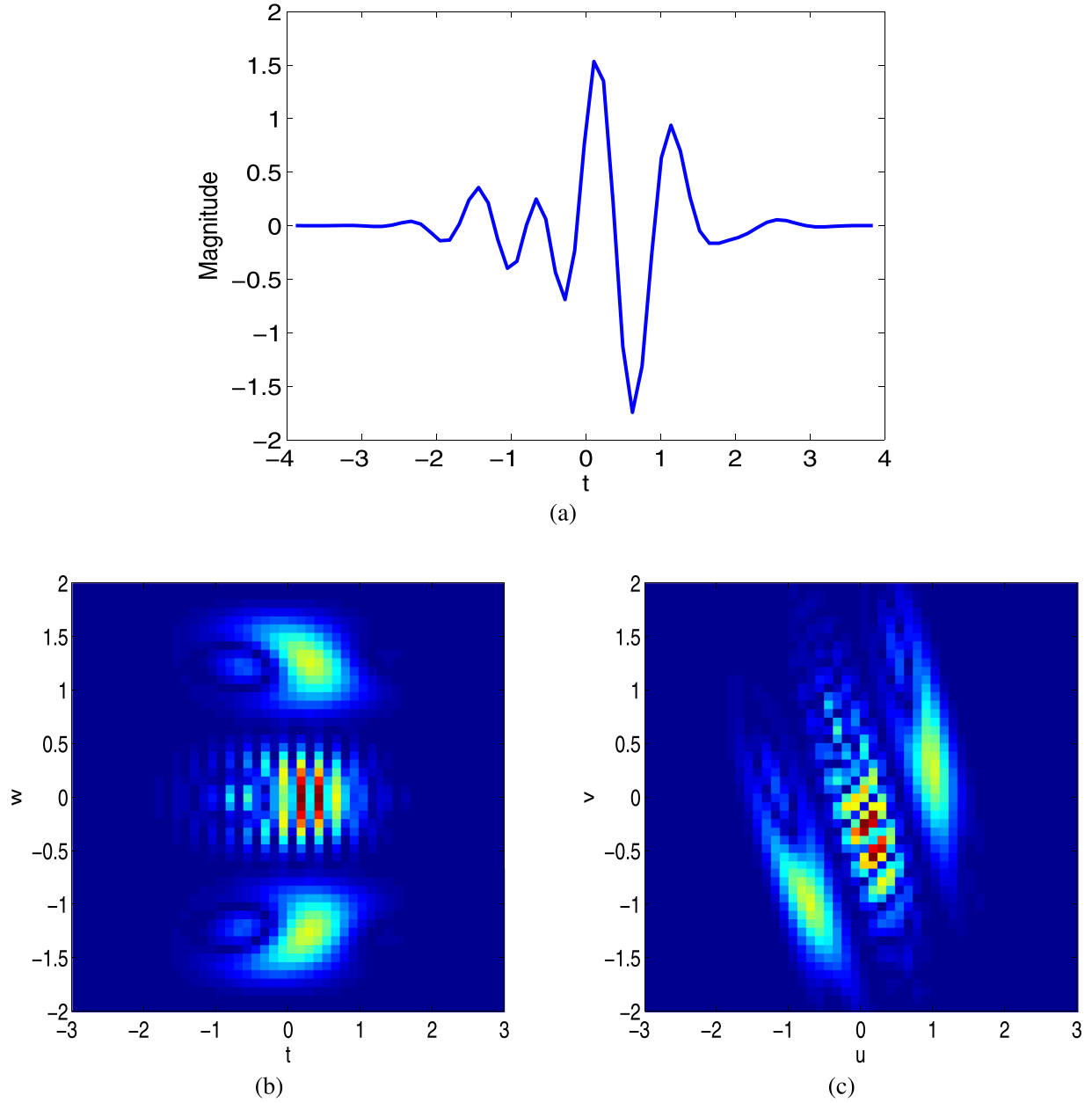


Fig. 4. (a) The original signal $s(t)$; (b) Wigner-Ville distribution (WVD) of original signal $s(t)$; (c) WVD of the transformed signal after the LCT operation.

4.1. The first case

Based on the definitions of the LCT and $z1_{A,\phi}(t)$, the spectrum of the first ϕ -LCAS in the LCT domain is

$$\begin{aligned}\mathcal{L}_{z1_{A,\phi}}^A(u) &= \int_{-\infty}^{+\infty} e^{-j\frac{u^2}{2b}} \tilde{z}_\phi(t) K_A(u, t) dt \\ &= \frac{e^{-j\phi}}{\sqrt{2j\pi b}} e^{j\frac{u^2}{2b}} 2U(u) S\left(\frac{u}{b}\right),\end{aligned}\quad (27)$$

where $S(\frac{u}{b})$ is the FT of $s(t)$. Eq. (27) shows that $z1_{A,\phi}(t)$ contains only the positive frequency in the LCT domain.

In order to judge whether this case is the eigenfunction of the ϕ -LCHT operator, we obtain

$$\begin{aligned}\mathcal{H}_{A,\phi}[z1_{A,\phi}](t) &= \cos\phi z1_{A,\phi}(t) + \sin\phi \mathcal{H}_A[z1_{A,\phi}](t) \\ &= e^{-j\phi} z1_{A,\phi}(t).\end{aligned}\quad (28)$$

Eq. (28) shows that $z1_{A,\phi}$ is indeed the eigenfunction of the ϕ -LCHT operator, and the eigenvalue is $e^{-j\phi}$, which is related to the rotation parameter ϕ .

4.2. The second case

In order to derive the LCT spectrum of $z2_{A,\phi}(t)$, the second ϕ -LCAS can first be simplified as

$$\begin{aligned}z2_{A,\phi}(t) &= s(t) + e^{j(\pi-\phi)} (\cos\phi s(t) + \sin\phi \mathcal{H}_A[s](t)) \\ &= j \sin\phi e^{-j\phi} z_A(t).\end{aligned}\quad (29)$$

From Eq. (29), $z2_{A,\phi}(t)$ is a scale of the GAS, and the scalar is related to the rotation parameter ϕ . Then, the LCT of $z2_{A,\phi}(t)$ is easily obtained by

$$\mathcal{L}_{z2_{A,\phi}}^A(u) = j \sin\phi e^{-j\phi} \mathcal{L}_{z_A}^A(u) = j \sin\phi e^{-j\phi} 2U(u) S^A(u). \quad (30)$$

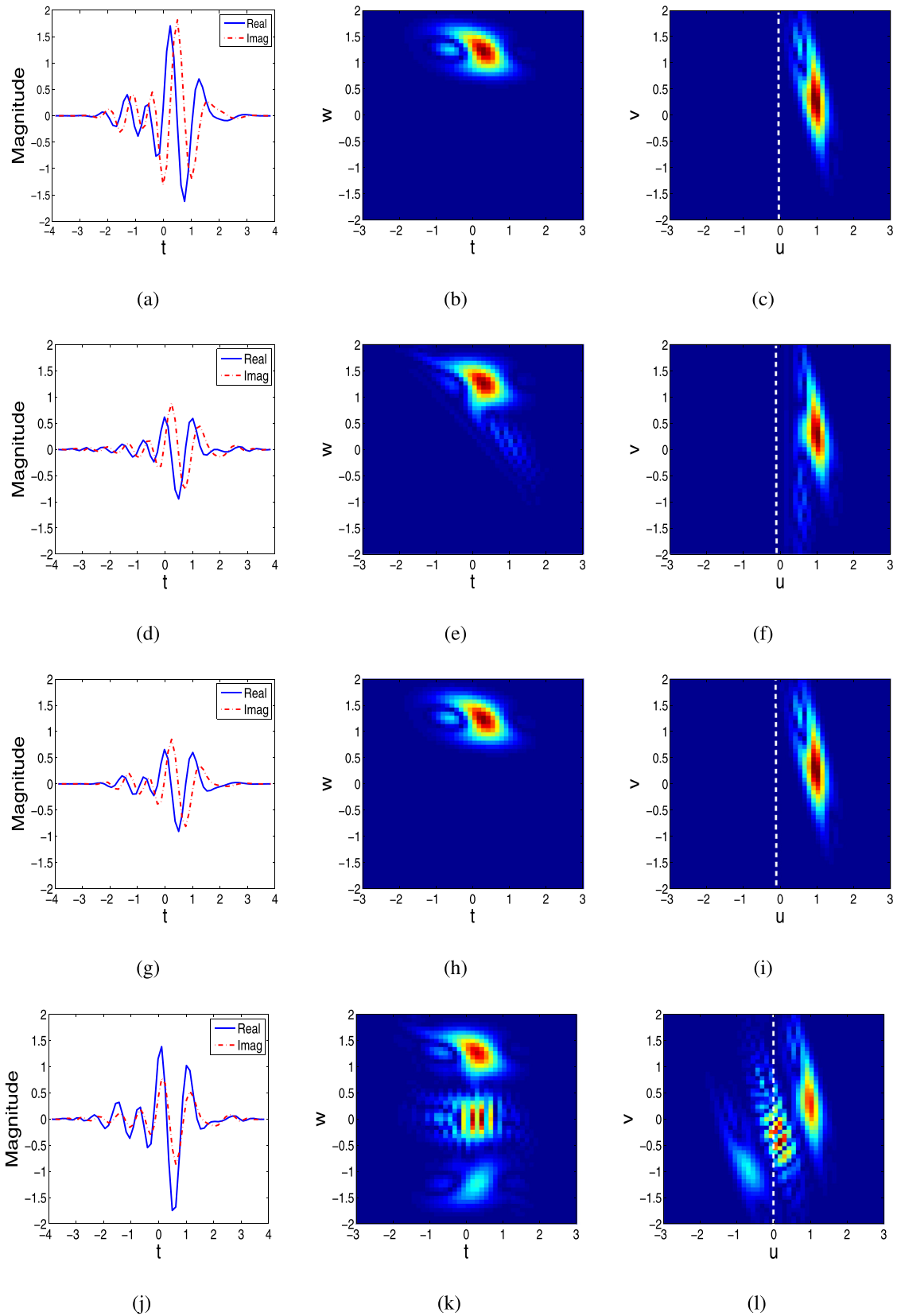


Fig. 5. ϕ -LCAS waveform, WVD, and WVD of the signals after the LCT with parameter matrix A_1 : (a–c) the first case; (d–f) the second case; (g–i) the third case; (j–l) the fourth case.

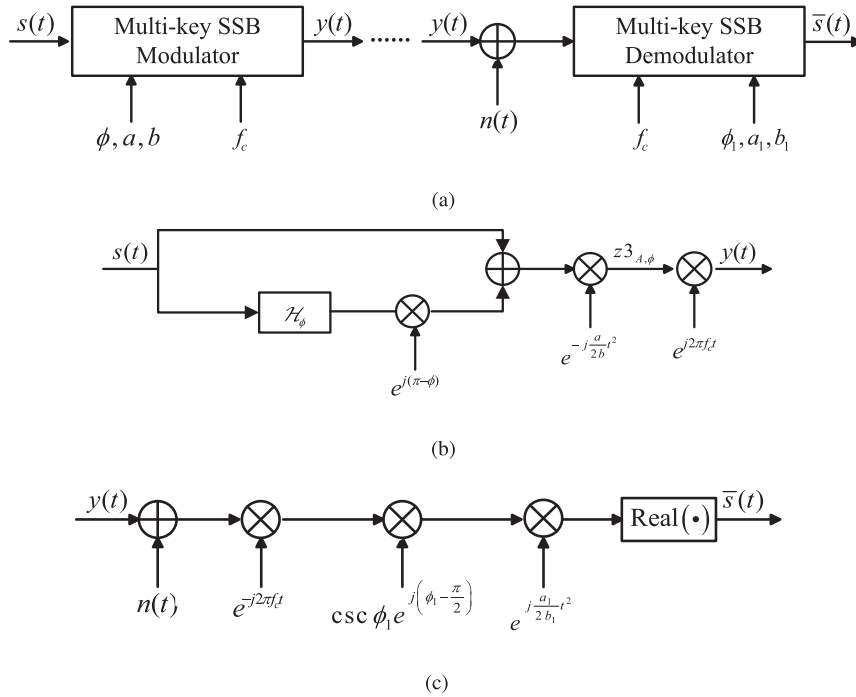


Fig. 6. (a) Multi-key secure SSB modulation scheme; (b) details of modulator; (c) details of demodulator.

Similar to Eq. (28), we derive

$$\begin{aligned}\mathcal{H}_{A,\phi}[z_{2A,\phi}](t) &= \cos \phi z_{2A,\phi}(t) + j \sin^2 \phi e^{-j\phi} \mathcal{H}_A[z_A](t) \\ &= e^{-j\phi} z_{2A,\phi}(t).\end{aligned}\quad (31)$$

The result in Eq. (31) shows that $z_{2A,\phi}$ is also the eigenfunction of the ϕ -LCHT operator with the same eigenvalue of $z_{1A,\phi}$.

4.3. The third case

By simple calculations, $z_{3A,\phi}(t)$ is equal to

$$\begin{aligned}z_{3A,\phi}(t) &= e^{-j\frac{a}{2b}t^2} [j \sin \phi e^{-j\phi} s(t) - \sin \phi e^{-j\phi} \mathcal{H}[s](t)] \\ &= j \sin \phi z_{1A,\phi}(t).\end{aligned}\quad (32)$$

The LCT of $z_{3A,\phi}(t)$ is

$$\mathcal{L}_{z_{3A,\phi}}(u) = j \sin \phi \frac{e^{-j\phi}}{\sqrt{2j\pi b}} e^{j\frac{bu^2}{2b}} 2U(u) S\left(\frac{u}{b}\right).\quad (33)$$

The eigenfunction of the ϕ -LCHT operator is

$$\mathcal{H}_{A,\phi}[z_{3A,\phi}](t) = j \sin \phi \mathcal{H}_{A,\phi}[z_{1A,\phi}](t) = e^{-j\phi} z_{3A,\phi}(t).\quad (34)$$

In particular, when $a = \cos \alpha$ and $b = \sin \alpha$, this kind of ϕ -LCAS reduces to that of Tao et al. [28].

4.4. The fourth case

In this case, the LCT of $z_{4A,\phi}(t)$ is

$$\begin{aligned}\mathcal{L}_{z_{4A,\phi}}^A(u) &= \cos \phi S^A(u) + j \sin \phi \mathcal{L}_{\mathcal{H}_A[s]}^A(u) \\ &= (\cos \phi + \sin \phi \operatorname{sgn}(u)) S^A(u).\end{aligned}\quad (35)$$

In terms of the eigenfunction of the ϕ -LCHT operator,

$$\begin{aligned}\mathcal{H}_{A,\phi}[z_{4A,\phi}](t) &= \cos^2 \phi s(t) + j \cos \phi \sin \phi \mathcal{H}_A[s](t) + \sin \phi \cos \phi \mathcal{H}_A[s](t) \\ &\quad - j \sin^2 \phi s(t) \\ &= (\cos^2 \phi - j \sin^2 \phi) s(t) + (1 + j) \sin \phi \cos \phi \mathcal{H}_A[s](t).\end{aligned}\quad (36)$$

From Eqs. (35) and (36), the LCT spectrum and eigenfunction properties of the ϕ -LCHT operator are related to ϕ but not preserved.

5. The merits of ϕ -linear canonical analytic signals

In order to better explain the characteristics of the four different ϕ -LCASs, their merits are summarized and discussed using examples in the time-frequency plane.

First, we make several important remarks about the four ϕ -LCASs:

- (M1) From the LCT spectrum, $z_{1A,\phi}(t)$, $z_{2A,\phi}(t)$, and $z_{3A,\phi}(t)$ maintain the one-sided spectrum in the LCT domain. However, $z_{4A,\phi}(t)$ cannot preserve this property; that is, $z_{4A,\phi}(t)$ contains negative contents of the LCT spectrum unlike $z_{1A,\phi}(t)$, $z_{2A,\phi}(t)$, and $z_{3A,\phi}(t)$ (see Eqs. (27), (30), (33), and (35)).
- (M2) From the eigenfunction property, the first three ϕ -LCASs are the eigenfunctions of the ϕ -LCHT operator according to Eqs. (28), (31), and (34). Moreover, they all belong to the same eigenvalue, $e^{-j\phi}$.
- (M3) From the SSB application, because of (M1), the possible tools used in the SSB are $z_{1A,\phi}(t)$, $z_{2A,\phi}(t)$, and $z_{3A,\phi}(t)$. However, the one-sided spectrum of $z_{2A,\phi}(t)$ is the LCT of $s(t)$. This analytic version cannot contain the exact same information as $s(t)$ after suppressing the spectrum components by half in the LCT domain [16]. While $z_{1A,\phi}(t)$ and $z_{3A,\phi}(t)$ are Fourier spectra and $e^{-j\frac{a}{2b}t^2}$ is a symmetric function of u , these two ϕ -LCASs can be used in the SSB scheme.

The widely used time-frequency analysis tool, specifically, the WVD, is applied to facilitate understanding of the LCT spectrum property of the four ϕ -LCASs in the time-frequency domain. More details about the relation between the LCT and WVD can be found in [29]. Here, $s(t)$ is a real valued, Gaussian-weighted sinusoidal signal given by

$$s(t) = \cos(2\pi t) \cdot \exp(-(t - 0.3)^2) + \sin(3\pi t) \cdot \exp(-t^2).\quad (37)$$

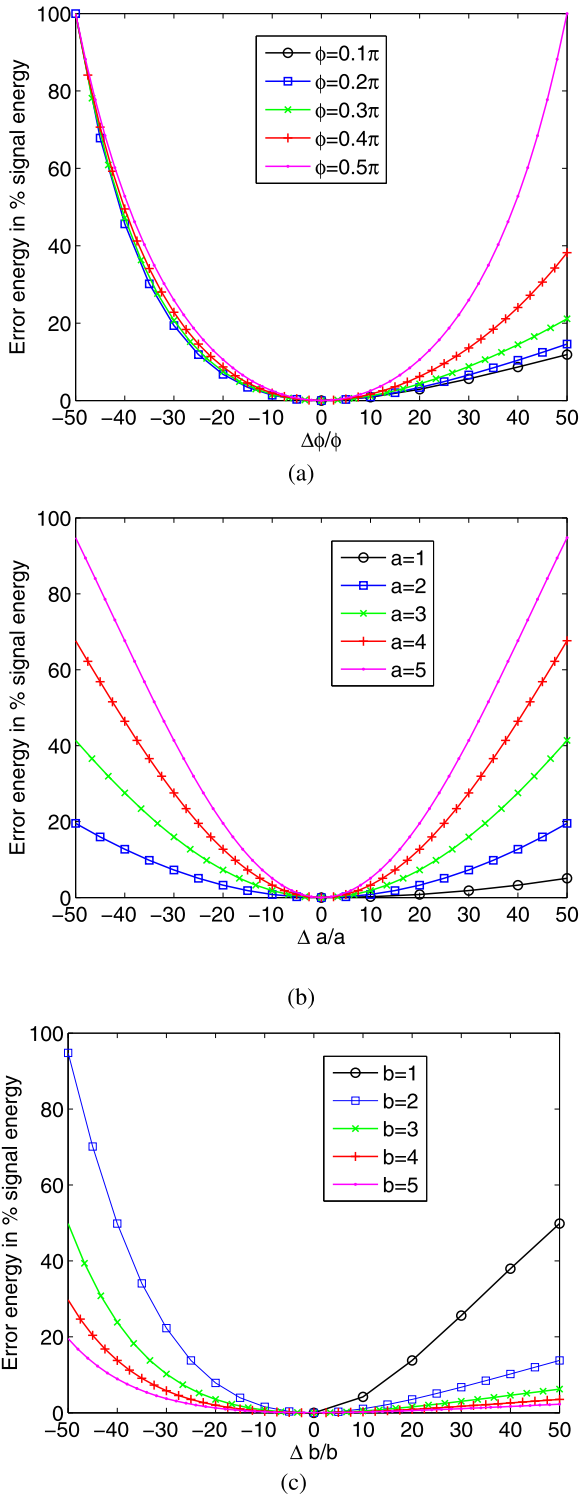


Fig. 7. Energy of the error versus key perturbation for different values of the security keys: (a) rotation parameter ϕ , (b) parameter a , (c) parameter b .

The sample interval is $1/\sqrt{N}$ and $N = 101$. The test LCT parameter matrix A_1 is an FrFT parameter case with $\alpha = \pi/3$, i.e., $A_1 = (\cos \alpha, \sin \alpha; -\sin \alpha, \cos \alpha) = (1/2, \sqrt{3}/2; -\sqrt{3}/2, 1/2)$. Fig. 4 shows the waveform of the original signal $s(t)$, the WVD of $s(t)$, and the WVD of the transformed signal after the LCT operator with parameter matrix A_1 . Because of the conjugate symmetry of the FT for the real signal $s(t)$, the WVD of $s(t)$ is symmetrical with respect to frequency w . Moreover, the WVD of the transformed

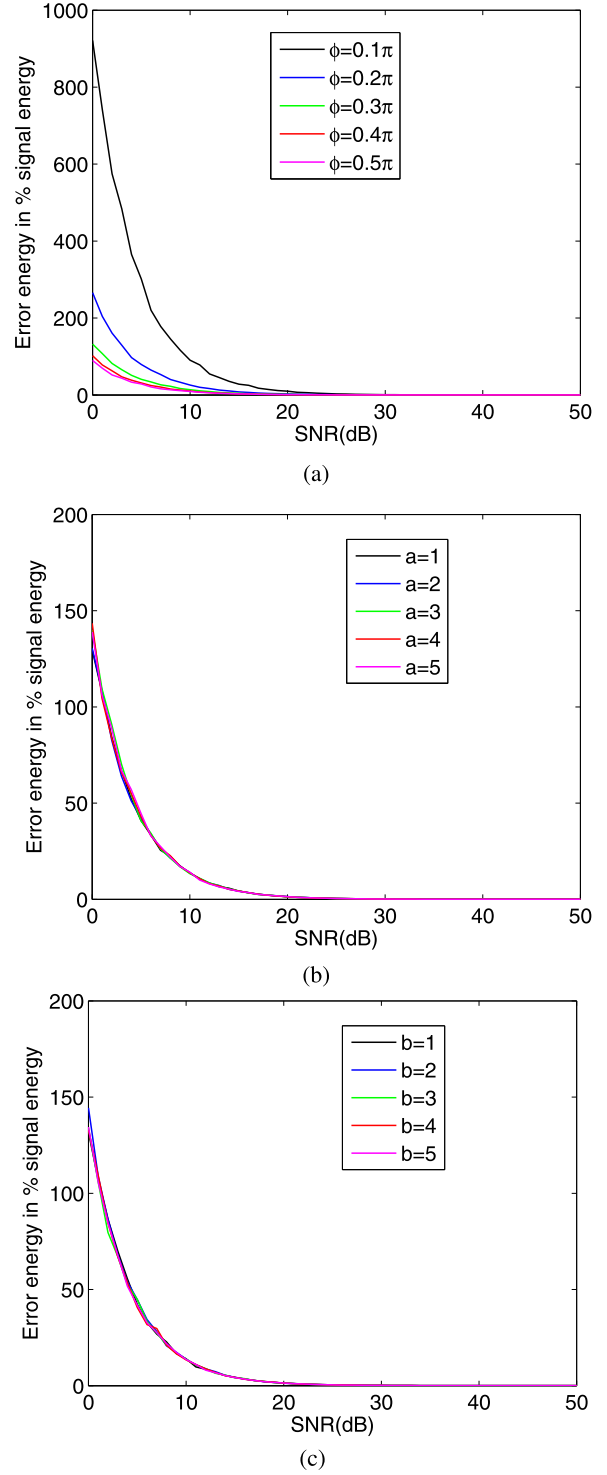


Fig. 8. Energy of the error as a function of the signal-noise-rate (SNR) for different values of security keys: (a) rotation parameter ϕ , (b) parameter a , (c) parameter b .

signal has a $\pi/3$ rotation of the WVD of the original signal. The WVD of the LCT-transformed signal is a two-dimensional linearly distorted version of the original WVD.

Fig. 5 shows the experimental results of the four ϕ -LCASs, including waveforms, the corresponding WVDs, and the WVDs of the transformed ϕ -LCASs after the LCT with parameter matrix A_1 . According to the definitions of ϕ -LCASs in Eq. (26), we demonstrate that their WVDs are quite different, as shown in Fig. 5(b), (e), (h), and (k). More specifically, in Fig. 5(b), (e), and (h), the spectra of

the ϕ -LCAS in the time-frequency domain are concentrated on the upper-half of the plane along the positive axis w . However, the spectrum of $z_{A_1, \phi}$ spreads along the whole axis w , as shown in Fig. 5(k). By observing the WVD of the transformed ϕ -LCASs, the one-sided LCT spectrum property can be found in Fig. 5(c), (f), and (i), but not in Fig. 5(l). That is, the spectra of the signals after the LCT are concentrated on the right half plane along the positive axis u in Fig. 5(c), (f), and (i), but not in Fig. 5(l), which corresponds to **M1**. Moreover, the WVDs of $z_{1A_1, \phi}$ and $z_{3A_1, \phi}$ contain complete positive frequency bands without distortion in Fig. 5(b), (h), (c), and (i) compared with the WVDs of the original signal $s(t)$ in Fig. 4(b) and (c), which relates to **M3**.

From the results of the above analysis, we have a comprehensive understanding of ϕ -LCASs. Thus, one can choose the proper case in practical applications according to their merits. For example, two points of generalized analytic signals are necessary in SSB applications: one is the one-sided spectrum property, and the other is to carry all of the information of the original signal. Both $z_{1A_1, \phi}$ and $z_{3A_1, \phi}$ are satisfied with these two conditions according to the above analysis. The only difference between $z_{1A_1, \phi}$ and $z_{3A_1, \phi}$ is a scaler related to the rotation parameter (see Eq. (32)); one of them is chosen to be used for an SSB modulation application in the next section.

6. Application of ϕ -linear canonical analytic signals

In this section, a ϕ -LCAS is applied to construct a secure SSB communication system. For the purpose of security, the AS in the LCT domain is obtained by the definition of $z_{A_1, \phi}$ with multiple secret keys a , b , and ϕ . The proposed SSB modulation system is shown in Fig. 6, where f_c is the carrier frequency. The signal $s(t)$ is encoded through a fractional Hilbert filter, multiplying a chip signal with parameters a and b to generate the ϕ -LCAS, which is transmitted after modulating with a carrier of frequency f_c . At the receiver, $y(t)$ is demodulated to obtain the ϕ -LCAS, which is further decoded using the rotation parameter ϕ_1 and parameters a_1 and b_1 . Let the signal $x(t)$ be defined by

$$x(t) = e^{j\frac{a_1}{2b_1}t^2} \cdot e^{j(\phi_1 - \frac{\pi}{2})} \csc \phi_1 z_{3A_1, \phi_1}(t). \quad (38)$$

When $a_1 = a$, $b_1 = b$, and $\phi_1 = \phi$,

$$s(t) = \text{Re}(x(t)). \quad (39)$$

Proof. Substituting $z_{3A_1, \phi}(t) = e^{-j\frac{a}{2b}t^2} j \sin \phi e^{-j\phi} z(t)$ in Eq. (38), we obtain

$$x(t) = e^{j\frac{a}{2b}t^2} \cdot e^{j(\phi - \frac{\pi}{2})} \csc \phi z_{3A_1, \phi} = z(t). \quad (40)$$

Based on the relation between a given real signal $s(t)$ and its AS $z(t)$ in Eq. (2), i.e., $s(t)$ is exactly the real part of $z(t)$, the recovery of input $s(t)$ is from the real part of Eq. (38) according to Eq. (40). From Eqs. (38) and (39), the decoded signal (denoted by $\bar{s}(t)$) will differ from the desired signal $s(t)$ unless the keys used at the receiver end coincide exactly with that employed for encoding, i.e., $\Delta\phi = 0$, $\Delta a = 0$, and $\Delta b = 0$. The first problem discussed below is the output of the demodulator when parameters a_1 , b_1 , and ϕ_1 in the demodulator are not equal to the parameters in the modulator, that is, the effect of key perturbations.

We use the energy of the error $E_{(s-\bar{s})}$ between $s(t)$ and the recovery signal $\bar{s}(t)$ as an index to analyze the effect of key perturbations under the noiseless condition. Here, the signal $s(t)$ is given by $s(t) = (1 + 0.5 \cos(10\pi t)) \cdot \cos(50\pi t)$ for the interval $[-1, 1]$ and sampling frequency $f_s = 1000$ Hz. Fig. 7 shows $E_{(s-\bar{s})}$ as a function of $\Delta\phi/\phi \times 100$, $\Delta a/a \times 100$ and $\Delta b/b \times 100$, expressed as a percentage of the signal energy E_s . Notice that the error is zero for all three parameters when the perturbation is zero. Furthermore, for

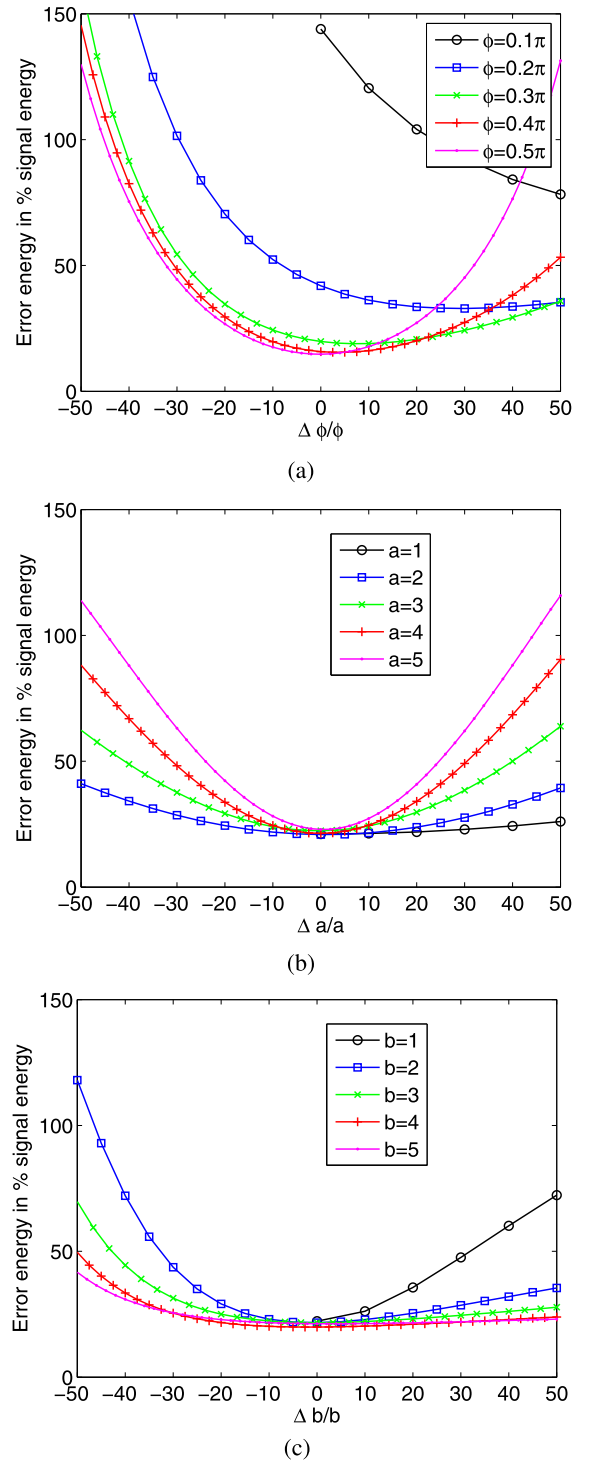


Fig. 9. Energy of the error versus key perturbation under the noise condition for different values of the security keys: (a) rotation parameter ϕ , (b) parameter a , (c) parameter b .

a given perturbation of parameters ϕ and a , the larger the magnitude of the encoder key, the larger the error. For parameter b , the opposite is observed.

In fact, SSB systems are often disturbed by noise in practical applications. It is important to analyze the performance of our proposed SSB modulation scheme under the noisy condition. The second problem discussed below is the effect of noise. Here, additive Gaussian white noise $n(t)$ is regarded as channel noise. The exper-

iment is performed under different signal-noise-rate (SNR) levels with no errors of key perturbations in Fig. 8. Notice that the energy of the error decreases as the SNR increases. Also notice that in the case of Fig. 8(b) and (c), the noise of the performance coincides for different key values, which is quite different from Fig. 8(a). This characteristic can be explained as follows. In the noisy case, the output of the demodulator becomes $\tilde{s}(t) = s(t) + u(t)$, where $u(t)$ represents the noise term given by

$$u(t) = \csc \phi \operatorname{Re} \left(e^{j \frac{a}{2b} t^2} \cdot e^{j(\phi - \frac{\pi}{2})} e^{-j2\pi f_c t} n(t) \right). \quad (41)$$

From Eq. (41), if parameter ϕ approaches zero, $\csc \phi$ approaches infinity. The error becomes very different because of the characteristics of $\csc \phi$. For parameters a and b , the error is stable since a and b are located in the real part of the exponential function. Furthermore, the performance of key perturbations under a given SNR condition (SNR = 8 dB) is shown in Fig. 9. Compared with the three cases, the SSB system becomes quite stable under a and b , which is contrary to the stability for ϕ .

Based on the above analysis, proper keys a , b , and ϕ can be set according to their characteristics. Once the keys are not matched, the demodulator will have a large error. Therefore, multi-keys effectively improve security performance of the SSB. In addition, because of parameter b in the LCT spectrum of the ϕ -LCAS, we are free to choose a better degree of bandwidth.

7. Conclusions

In this paper, we formulated a new extension of the HT based on the LCT and rotation parameter; specifically, we formulated the ϕ -LCHT. The ϕ -LCHT is a linear combination of the original signal and its PHT, including many generalized HTs as special cases. It satisfies essential properties that earlier definitions could not satisfy. Interpretations of this new definition were discussed from geometry and filter function views. According to the meanings of parameters, four ϕ -LCASs were derived from different perspectives and merits were compared in detail using the WVD. The performances of the ϕ -LCASs in an SSB modulation application were discussed in detail considering the effects of key perturbations and noise. We verified that the ϕ -LCAS can improve the security of SSB modulation using multi-keys. This approach leads to a further generalization of the HT/AS and offers many more options for practical applications.

Acknowledgements

This work is supported by the National Natural Science Foundation of China (No. 61671063), and also by the Foundation for Innovative Research Groups of the National Natural Science Foundation of China (No. 61421001).

References

- [1] D. Gabor, Theory of communication, J. Inst. Elect. Eng. 93 (1946) 429–457.
- [2] J.A. Davis, D.E. McNamara, D.M. Cottrell, Analysis of the fractional Hilbert transform, Appl. Opt. 37 (29) (1998) 6911–6913.
- [3] B. Boashash, Estimating and interpreting the instantaneous frequency of a signal-part 1: fundamentals, Proc. IEEE 80 (4) (1992) 520–538.
- [4] B. Boashash, Estimating and interpreting the instantaneous frequency of a signal-part 2: algorithms and applications, Proc. IEEE 80 (4) (1992) 540–568.
- [5] S.M. Kay, Maximum entropy spectral estimation using the analytical signal, IEEE Trans. Acoust. Speech Signal Process. 26 (5) (1978) 467–469.
- [6] A.I. Zayed, Advances in Shannon's Sampling Theory, CRC, Boca Raton, 1993.
- [7] S.L. Marple, Estimating group delay and phase delay via discrete-time analytic cross-correlation, IEEE Trans. Signal Process. 47 (9) (1999) 2604–2607.
- [8] R.C. Cabot, A note on the application of the Hilbert transform to time delay estimation, IEEE Trans. Acoust. Speech Signal Process. 29 (3) (1981) 607–609.
- [9] I. Daubechies, The wavelet transform, time-frequency localization and signal analysis, IEEE Trans. Inf. Theory 36 (5) (1990) 961–1005.
- [10] H.M. Ozaktas, M.A. Kutay, Z. Zalevsky, The Fractional Fourier Transform with Applications in Optics and Signal Processing, John Wiley & Sons, Chichester, New York, USA, 2001.
- [11] A. Koc, H.M. Ozaktas, C. Candan, M.A. Kutay, Digital computation of linear canonical transforms, IEEE Trans. Signal Process. 56 (6) (2008) 2383–2394.
- [12] T.Z. Xu, B.Z. Li, Linear Canonical Transform and Its Applications, Science Press, Beijing, 2013.
- [13] A.I. Zayed, Hilbert transform associated with the fractional Fourier transform, IEEE Signal Process. Lett. 5 (8) (1998) 206–208.
- [14] Y.X. Fu, L.Q. Li, Generalized analytic signal associated with linear canonical transform, Opt. Commun. 281 (6) (2008) 1468–1472.
- [15] Y.X. Fu, L.Q. Li, A generalized Bedrosian theorem in fractional Fourier domain, in: International Conference on Computational Intelligence and Security (CIS, 2006), 2006, pp. 1785–1788.
- [16] S.C. Pei, Y.C. Lai, Derivation and discrete implementation for analytic signal of linear canonical transform, J. Opt. Soc. Am. A 30 (5) (2013) 987–992.
- [17] G.L. Xu, X.T. Wang, X.G. Xu, Generalized Hilbert transform and its properties in 2D LCT domain, Signal Process. 89 (7) (2009) 1395–1402.
- [18] G.L. Xu, X.T. Wang, L.T. Wang, B. Liu, S.P. Su, X.G. Xu, Generalized uncertainty principles associated with Hilbert transform, Signal Image Video Process. 8 (2) (2014) 279–285.
- [19] C.C. Tseng, S.C. Pei, Design and application of discrete-time fractional Hilbert transformer, IEEE Trans. Circuits Syst. II Analog Digit. Signal Process. 47 (12) (2000) 1529–1533.
- [20] S.C. Pei, M.H. Yeh, Discrete fractional Hilbert transform, IEEE Trans. Circuits Syst. II Analog Digit. Signal Process. 47 (11) (2000) 1307–1311.
- [21] S.C. Pei, S.G. Huang, Reversible joint Hilbert and linear canonical transform without distortion, IEEE Trans. Signal Process. 61 (19) (2013) 4768–4781.
- [22] A.W. Lohmann, D. Mendlovic, Z. Zalevsky, Fractional Hilbert transform, Opt. Lett. 21 (4) (1996) 281–283.
- [23] A. Cusumariu, Fractional analytic signals, Signal Process. 82 (2) (2002) 267–272.
- [24] A. Venkitaraman, C.S. Seelamantula, Fractional Hilbert transform extensions and associated analytic signal construction, Signal Process. 94 (1) (2014) 359–372.
- [25] Y. Guo, B.Z. Li, Blind image watermarking method based on linear canonical wavelet transform and QR decomposition, IET Image Process. 10 (10) (2016) 773–786.
- [26] Q. Feng, B.Z. Li, Convolution and correlation theorems for the two-dimensional linear canonical transform and its applications, IET Signal Process. 10 (2) (2016) 125–132.
- [27] J.J. Healy, M.A. Kutay, H.M. Ozaktas, J.T. Sheridan, Linear Canonical Transforms: Theory and Applications, Springer, New York, 2016.
- [28] R. Tao, X.M. Li, Y. Wang, Generalization of the fractional Hilbert transform, IEEE Signal Process. Lett. 15 (24) (2008) 365–368.
- [29] S.C. Pei, J.J. Ding, Relations between fractional operations and time-frequency distributions, and their applications, IEEE Trans. Signal Process. 49 (8) (2001) 1638–1655.